Introduction to Python

In this lecture

- Data science
- Tools for data science
- History of Python
- Python IDEs

Introduction

- We live in a world that's drowning in data
- Data is generated from various sources
  - Websites track every user's every click
  - Your smartphone is building up a record of your location
  - Sensors from electronic devices record real time information
  - E-commerce websites collect purchasing habits

Data science

- Interdisciplinary field that brings together computer science, statistics and mathematics to extract useful insights from data

- Analyzing and generating insights from data aids in arriving at better business decisions

Popular tools used in data science

- Data pre-processing and analysis
  - Python, R, Microsoft Excel, SAS, SPSS
- Data exploration and visualization
  - Tableau, Qlikview, Microsoft Excel
- Parallel and distributed computing incase of big data
  - Apache Spark,Apache Hadoop

Evolution of Python

- Python was developed by Guido van Rossum in the late eighties at the 'National Research Institute for Mathematics and Computer Science' at Netherlands
- Python Editions
  - Python 1.0-1991,
  - Python 2.0- 2000
  - Python 3.0 - 2008 (Python 3.7 – latest)

Advantages of using python

- Python has several features that make it well suited for data science

- Open source and community development
  - Developed under Open Source Initiative approved license making it free to use and distribute even commercially

Advantages of using python

- Syntax used is simple to understand and code
- Libraries designed for specific data science tasks
- Combines well with majority of the cloud platform service providers

Integrated development environment (IDE)

- Software application consisting of a cohesive unit of tools required for development
- Designed to simplify software development
- Utilities provided by IDEs include tools for managing, compiling, deploying and debugging software

Features of IDE

- IDE should centralize three key tools that form the crux of software development
  - Source code editor
  - Compiler
  - Debugger
- Additional features
  - Syntax and error highlighting
  - Code completion
  - Version control

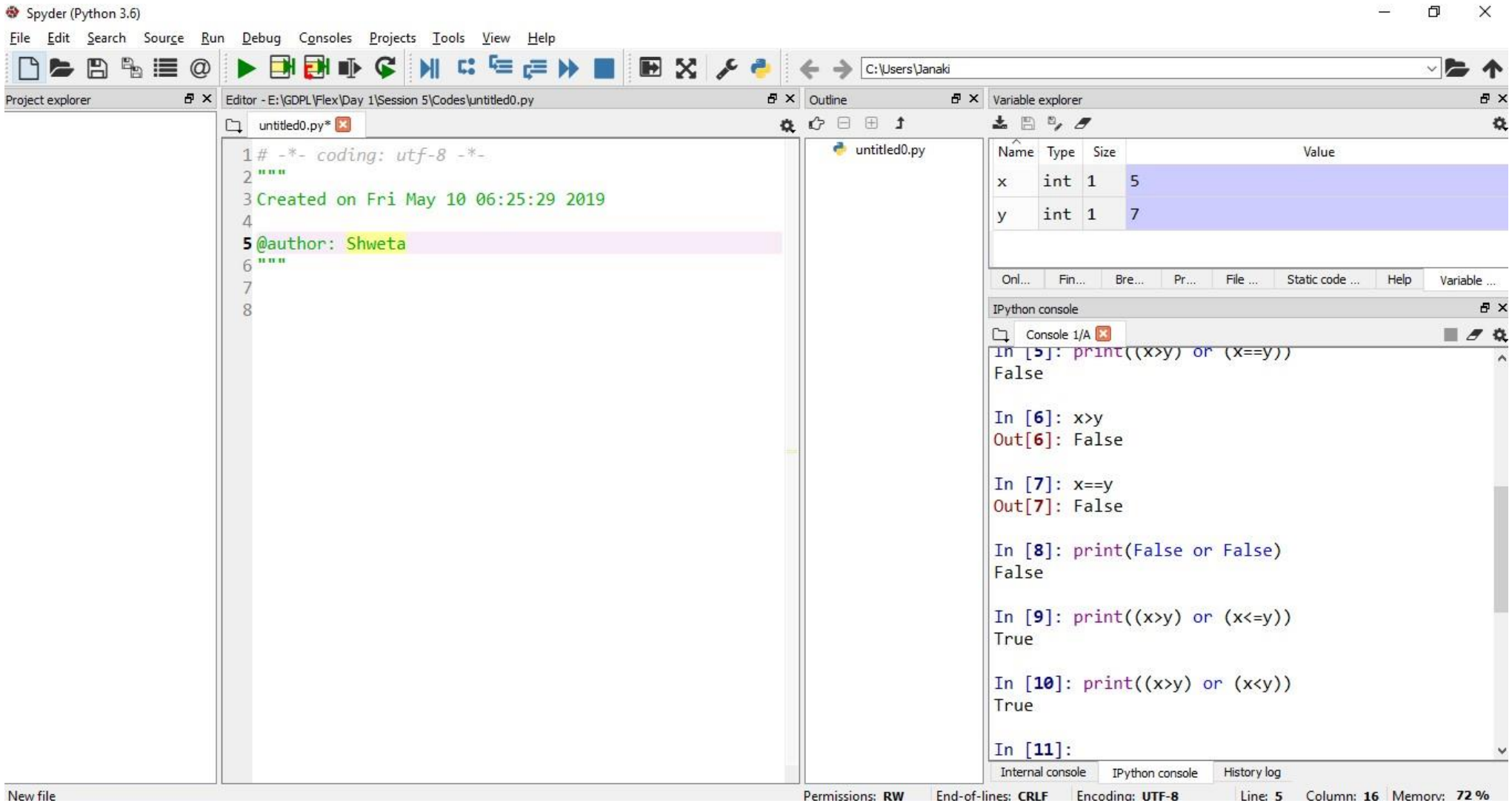
Commonly used IDEs

- Spyder
- PyCharm
- Jupyter Notebook
- Atom

Spyder

- Supported across Linux, Mac OS X and Windows platforms
- Available as open source version
- Bundled with Anaconda distribution which comes with all Python libraries
- Developed for Python and specifically data science

Spyder


Spyder (Python 3.6)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Janaki
Project explorer
Editor - E:\GDPL\Flex\Day 1\Session 5\Codes\untitled0.py
untitled0.py*
-*- coding: utf-8 -*-
"""
Created on Fri May 10 06:25:29 2019
@author: Shweta
"""
Outline
untitled0.py
Variable explorer
Name Type Size Value
x int 1 5
y int 1 7
Onl... Fin... Bre... Pr... File ... Static code ... Help Variable ...
IPython console
Console 1/A
In [5]: print((x>y) or (x==y))
False
In [6]: x>y
Out[6]: False
In [7]: x==y
Out[7]: False
In [8]: print(False or False)
False
In [9]: print((x>y) or (x<=y))
True
In [10]: print((x>y) or (x<y))
True
In [11]:
Internal console IPython console History log
New file
Permissions: RW End-of-lines: CRLF Encoding: UTF-8 Line: 5 Column: 16 Memory: 72 %

Spyder

- Features include
  - Code editor with robust syntax and error highlighting
  - Code completion and navigation
  - Debugger
  - Integrated document

- Interface similar to MATLAB and RStudio

PyCharm

- Supported across Linux, Mac OS X and Windows platforms
- Available as community (free open source) and professional (paid) version
- Supports only Python
- Bundled with Anaconda distribution which comes with all Python libraries
  - Can also be installed separately

PyCharm

Project1 [E:\GDPL\PyCharm\Projects\Project1] - ...\Problem_1\main_driver.py [Project1] - PyCharm

File Edit View Navigate Code Refactor Run Tools VCS Window Help

Project1 > Problem_1 > main_driver.py

main_driver

Project

- Project1 E:\GDPL\PyCharm\Projects\Project1
  - config
  - data
  - Problem_1
    - my_package
    - main_driver.py
  - requirements.txt
- External Libraries
  - < Python 3.7 (Project1) > E:\GDPL\PyCharm\Virtua
    - Binary Skeletons
    - DLLs
    - Extended Definitions
    - Lib
    - pip-10.0.1-py3.7.egg
    - Project1 library root
    - py4j-0.10.7-src.zip library root
    - python library root
    - Python37 library root
    - setuptools-39.1.0-py3.7.egg library root
    - site-packages
      - numpy
      - numpy-1.16.3.dist-info
      - pip-10.0.1-py3.7.egg
      - py4j
      - py4j-0.10.7.dist-info
      - easy-install.pth
      - setuptools.pth
      - setuptools-39.1.0-py3.7.egg
    - Typeshed Stubs

main_driver.py

```
# -*- coding: <utf-8> -*-
__author__ = "Shweta Sridhar"

if __name__ == "__main__":
    a=5
    b=6
    c=a*b

    print("Hello World!")
```

if __name__ == "__main__"

6: TODO Terminal Python Console Event Log

IDE and Plugin Updates: PyCharm is ready to update. (24 minutes ago)

9:1 CRLF UTF-8 4 spaces Python 3.7 (Project1)

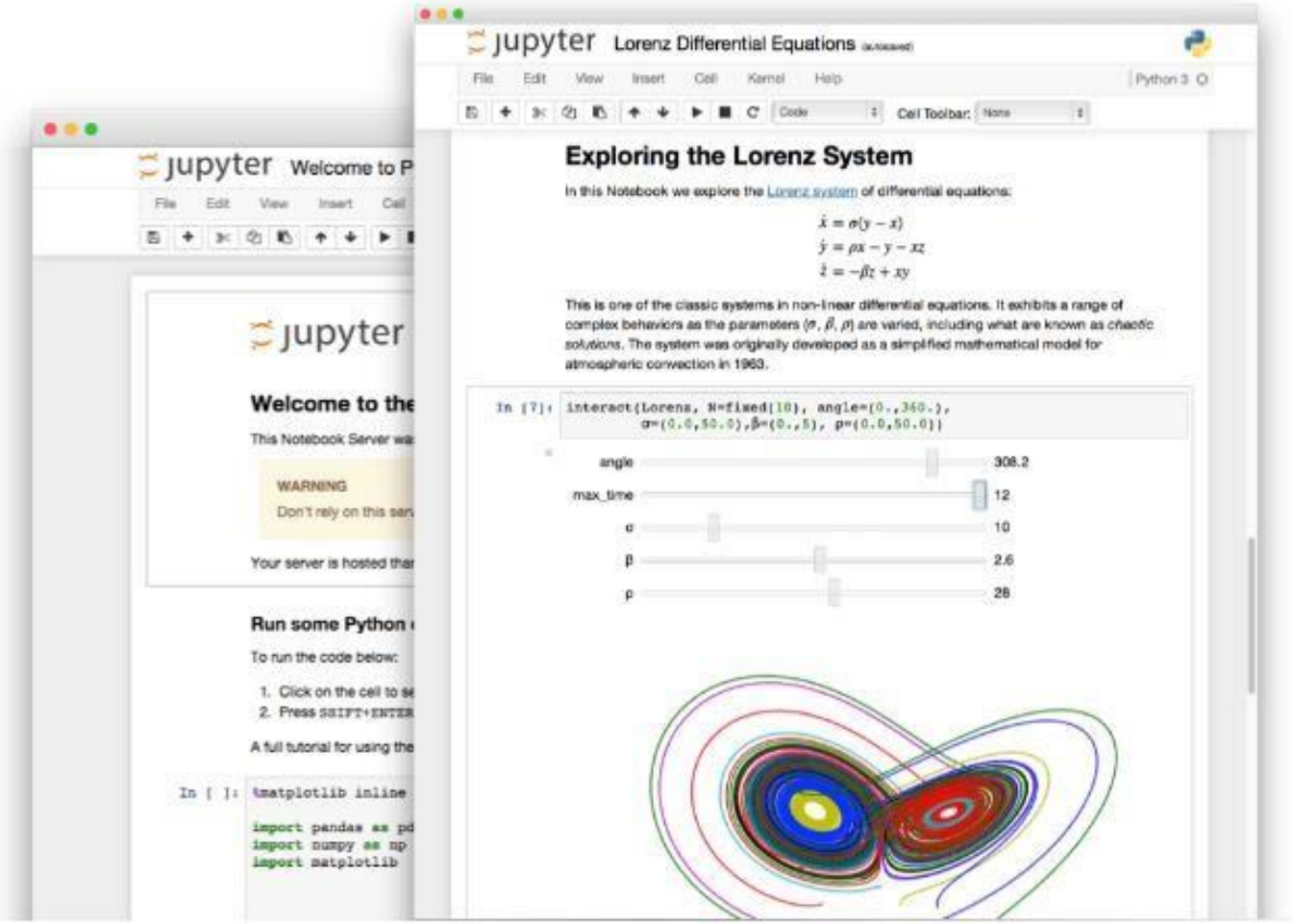
PyCharm

- Features include
  - Code editor provides syntax and error highlighting
  - Code completion and navigation
  - Unit testing
  - Debugger
  - Version control

Jupyter Notebook

- Web application that allows creation and manipulation of notebook documents called ‘notebook’
- Supported across Linux, Mac OS X and Windows platforms
- Available as open source version

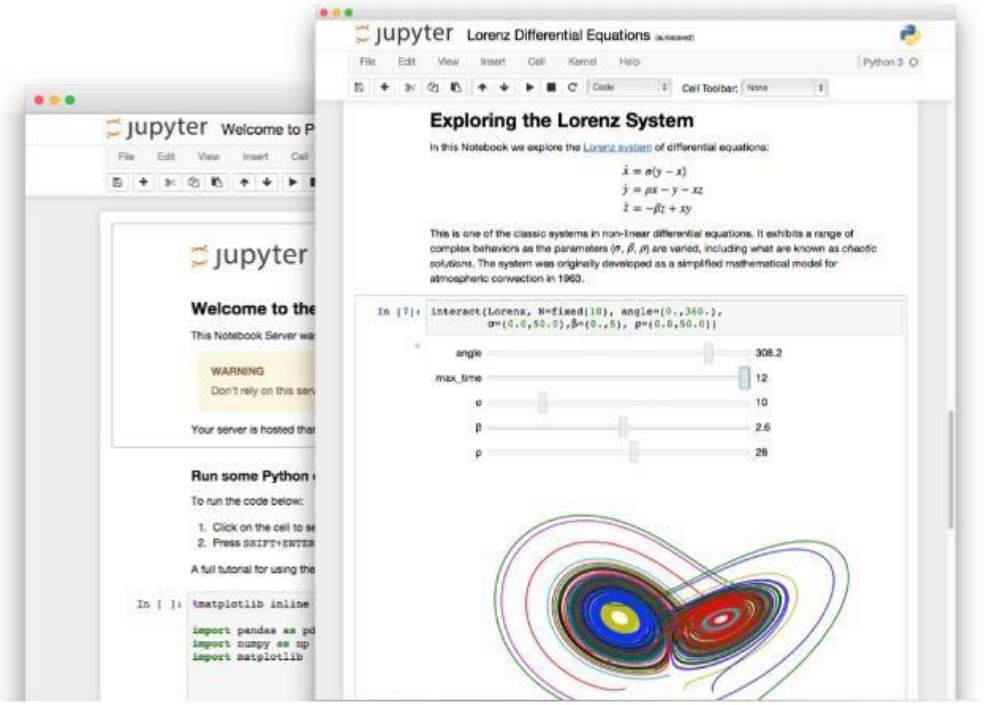
Jupyter Notebook



Source-https://jupyter.org/

Jupyter Notebook

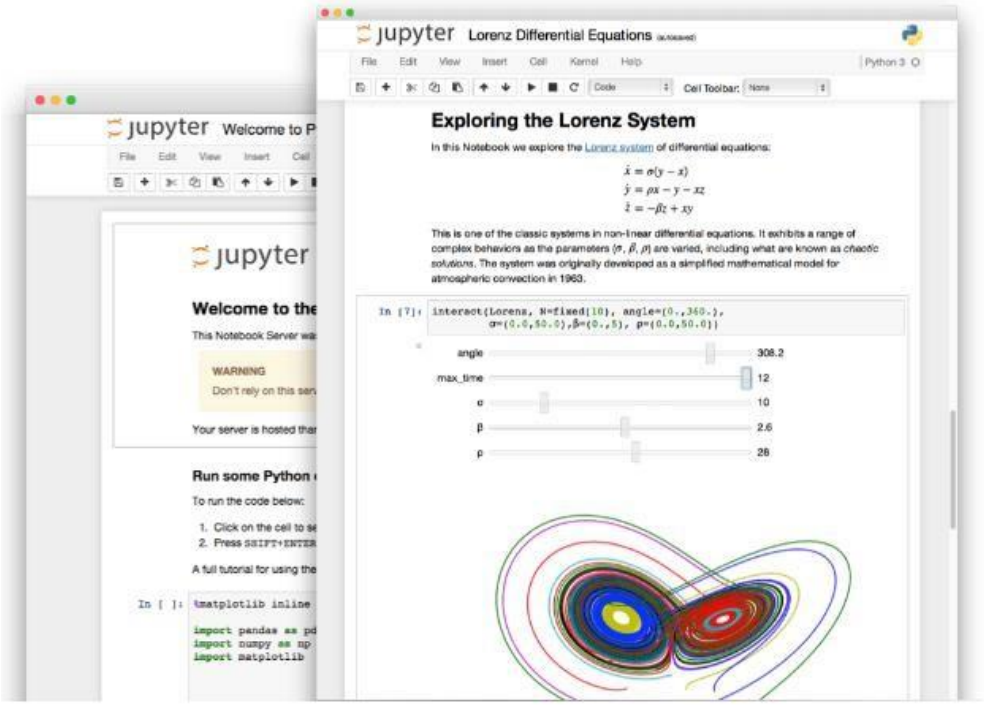
- Bundled with Anaconda distribution or can be installed separately
- Supports Julia, Python, R and Scala
- Consists of ordered collection of input and output cells that contain code, text, plots etc.



Source-https://jupyter.org/

Jupyter Notebook

- Allows sharing of code and narrative text through output formats like PDF, HTML etc.
  - Education and presentation tool
- Lacks most of the features of a good IDE

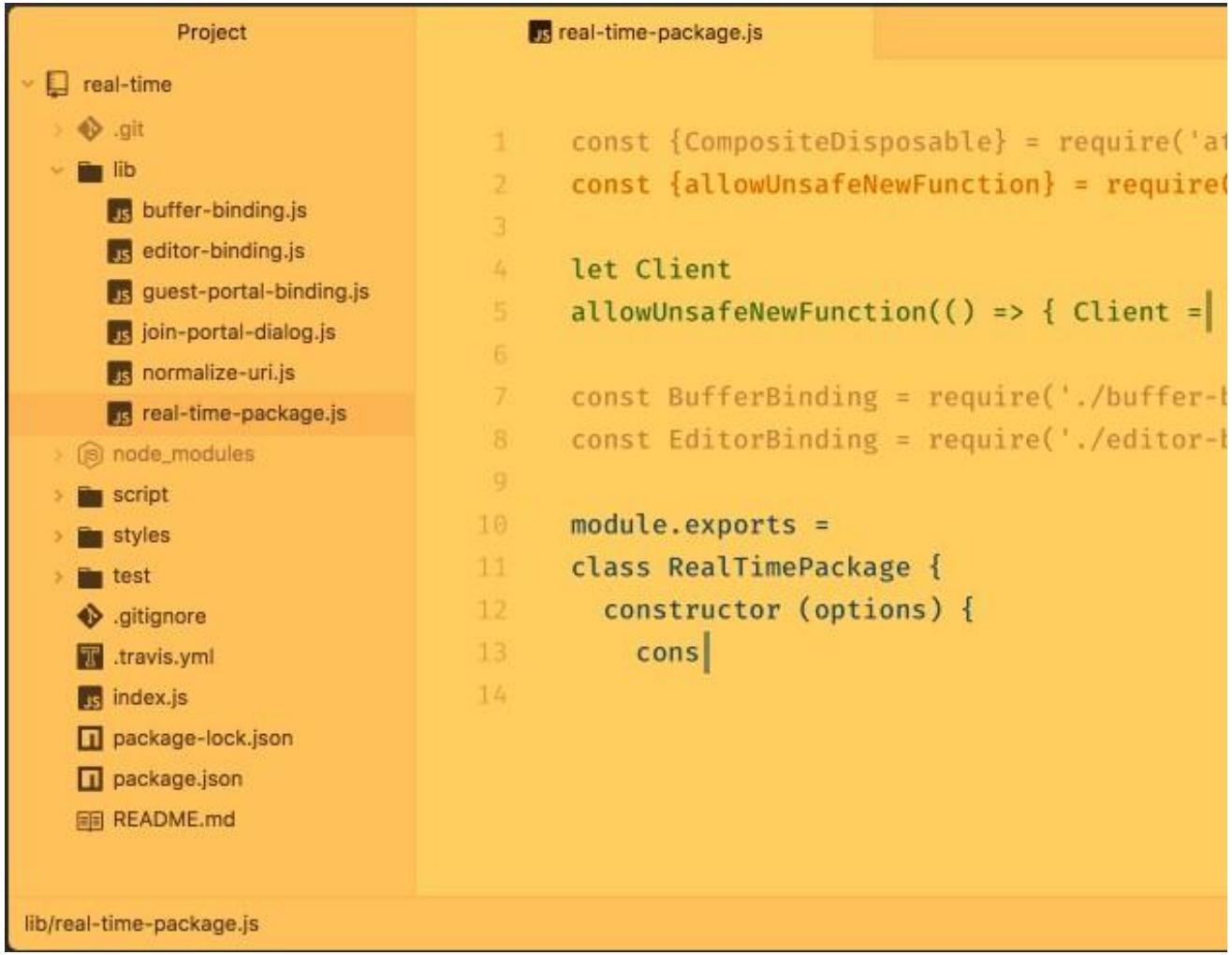


Source-https://jupyter.org/

Atom

- Open source text and source code editor
- Supported across Linux, Mac OS X and Windows platforms
- Supports Python, PHP, Java etc.
- Well suited for developers
- Enables users to install plug ins or packages
  - Packages can be installed for code completion, debugging

Atom



Source-https://atom.io/

How to choose the best IDE?

- Requirements
- Working with different IDEs helps us understand our own requirement
- In this course, Spyder will be used

Summary

- Popular tools used data science
- Evolution of Python
- Integrated development environment
  - Spyder
  - PyCharm
  - Jupyter Notebook
  - Atom

THANK YOU